

EXPLORATORY DATA ANALYSIS REPORT

BANKRUPTCY PREDICTION

Using Decision Trees and Random Forest with Precision–Recall Analysis

Machine Learning Project	B.Tech CSE A.Y. 2026–27
Dataset	UCI Taiwanese Bankruptcy Prediction
Dataset ID	572
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This report documents the exploratory analysis performed before model training. All numerical EDA results and visualizations were generated from the project dataset supplied for the bankruptcy prediction work.

Source: UCI Machine Learning Repository – Taiwanese Bankruptcy Prediction (Dataset ID 572), DOI 10.24432/C5004D.

ABSTRACT

Bankruptcy prediction is an important financial-risk problem because early identification of financial distress can support timely decision-making. This project uses supervised machine learning to classify companies as bankrupt or non-bankrupt using financial indicators. Before applying Decision Tree and Random Forest classifiers, Exploratory Data Analysis (EDA) is performed to understand data quality, class distribution and relationships between financial indicators and the bankruptcy target.

The dataset contains 6,819 company records and 95 financial features, together with the binary target variable “Bankrupt?”. The analysis checks dimensions, data types, missing values, duplicate records, descriptive statistics, constant features, class distribution and feature–target correlations. The EDA confirms that the target is strongly imbalanced: 6,599 observations are non-bankrupt and 220 are bankrupt.

The strongest associations with the bankruptcy target are concentrated around profitability, asset returns, net worth, leverage and liquidity-related indicators. These observations provide the evidence base for subsequent model development and explain why precision, recall, F1-score and Precision–Recall analysis are emphasized instead of relying on accuracy alone.

CHAPTER 1 – INTRODUCTION

1.1 Background

Companies can experience declining profitability, reduced liquidity and increasing financial obligations before bankruptcy. Traditional financial analysis often requires manual examination of multiple financial ratios. Machine learning provides a way to learn patterns from historical financial data and classify companies according to bankruptcy risk.

1.2 Purpose of Exploratory Data Analysis

EDA is performed before model training to understand the structure and quality of the dataset. It helps identify missing values, duplicate records, unusual or constant variables, class imbalance and relationships between financial indicators and the target variable. EDA also provides a basis for deciding which preprocessing and evaluation strategies are appropriate.

1.3 Objectives

1. Understand the dimensions and structure of the bankruptcy dataset.
2. Check missing values, duplicate records and feature data types.
3. Examine the distribution of bankrupt and non-bankrupt companies.
4. Identify constant or non-informative features.
5. Measure feature–target correlations and rank important indicators.
6. Compare important financial indicators across bankruptcy classes.
7. Provide an evidence-based foundation for Decision Tree and Random Forest modeling.

1.4 Scope

This report focuses on the exploratory stage of the project. Model training and final model performance are outside the main scope of this EDA report. The EDA findings are used to guide the subsequent supervised classification stage.

CHAPTER 2 – DATASET DESCRIPTION AND METHODOLOGY

2.1 Dataset Description

The project uses the Taiwanese Bankruptcy Prediction dataset from the UCI Machine Learning Repository. According to the official UCI documentation, the dataset contains 6,819 instances and 95 features. The data were collected from the Taiwan Economic Journal for the years 1999–2009, and company bankruptcy was defined according to Taiwan Stock Exchange business regulations.

Property	Value
Dataset	Taiwanese Bankruptcy Prediction
UCI Dataset ID	572
Instances	6,819
Financial features	95
Total columns in project CSV	96
Target variable	Bankrupt?

Task	Binary classification
Data period	1999–2009
Missing values (UCI documentation)	No
Project CSV missing values	0
Duplicate rows in project CSV	0
Constant feature detected	Net Income Flag

2.2 Feature Structure

The dataset contains financial indicators covering profitability, liquidity, leverage, operating performance, asset turnover, cash flow and growth-related measures. Examples include ROA measures, current and quick ratios, liability-to-equity measures, cash-flow ratios, net-income ratios and growth indicators. The target variable is an integer class label where 0 represents non-bankrupt and 1 represents bankrupt.

2.3 EDA Workflow

1. Load the CSV dataset into a Pandas DataFrame.
2. Inspect the first records, dimensions and column names.
3. Check data types and total missing values.
4. Check duplicate records.
5. Generate descriptive statistics.
6. Analyze the target-class distribution.
7. Detect constant features.
8. Replace infinite values with missing values for correlation calculations.
9. Calculate and rank feature–target correlations.
10. Visualize the strongest correlations with a bar chart and heatmap.
11. Compare the four strongest indicators using class-wise boxplots.

2.4 Tools and Technologies

Python, Pandas, NumPy, Matplotlib and Seaborn are used for data loading, statistical analysis and visualization. The same EDA workflow is implemented in the project Python program and can be executed in Google Colab or Jupyter Notebook.

CHAPTER 3 – EXPLORATORY DATA ANALYSIS

3.1 Dataset Loading and Dimensions

The project CSV was successfully loaded into a Pandas DataFrame. The resulting dataset contains 6,819 rows and 96 columns. Therefore, there are 95 financial predictor variables and one binary target variable.

3.2 Data Quality Analysis

Check	Result	Interpretation
Dataset shape	6,819 × 96	Large multivariate tabular dataset
Missing values	0	No missing cells detected
Duplicate rows	0	No duplicate records detected
Constant features	1	One non-varying feature was detected
Target classes	0 and 1	Binary classification problem

The constant feature identified by the analysis is “Net Income Flag”. Because a constant variable has the same value for every observation, it does not provide discriminative information and is excluded from meaningful feature analysis/model input.

3.3 Target Variable Analysis

The target variable is “Bankrupt?”. Class 0 contains 6,599 companies (96.77%), while class 1 contains 220 companies (3.23%). The difference demonstrates substantial class imbalance.

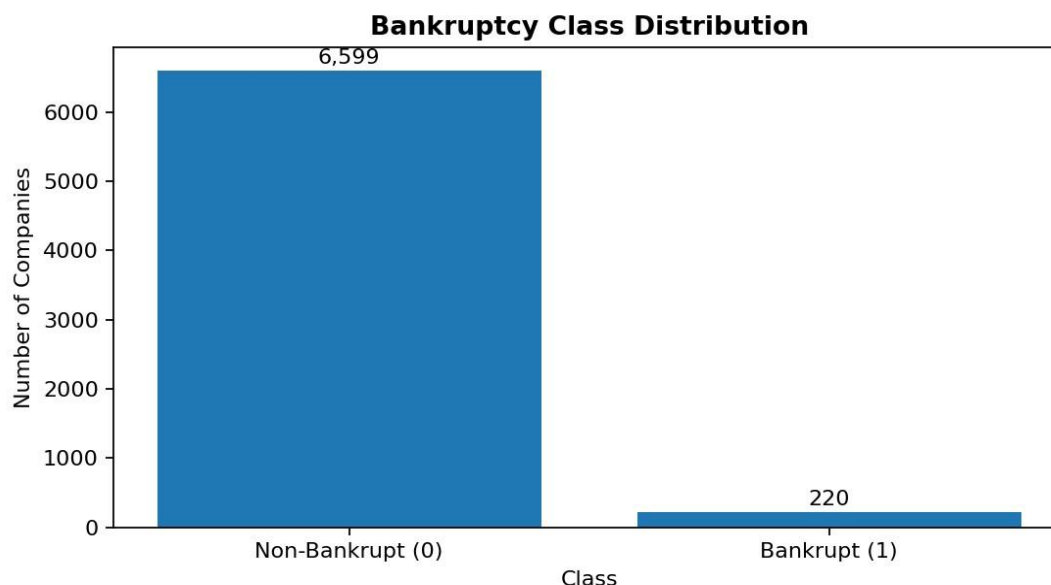


Figure 1. Distribution of non-bankrupt and bankrupt companies.

The strong imbalance is important for the modeling stage. A model that mostly predicts the majority class can achieve high accuracy while failing to detect bankrupt companies. This is why the project emphasizes Precision, Recall, F1-score and Precision–Recall analysis.

3.4 Descriptive Statistics

The dataset contains numerical financial indicators. Descriptive statistics were inspected to understand their central tendency and spread. The table below summarizes the ten strongest features by absolute correlation with the target.

Feature	Mean	Std. Dev.	Min	Max
Net Income to Total Assets	0.8078	0.0403	0.0000	1.0000
ROA(A) before interest and % after tax	0.5586	0.0656	0.0000	1.0000
ROA(B) before interest and depreciation after tax	0.5536	0.0616	0.0000	1.0000
ROA(C) before interest and depreciation before interest	0.5052	0.0607	0.0000	1.0000
Net worth/Assets	0.8868	0.0539	0.0000	1.0000
Debt ratio %	0.1132	0.0539	0.0000	1.0000
Persistent EPS in the Last Four Seasons	0.2288	0.0333	0.0000	1.0000
Retained Earnings to Total Assets	0.9347	0.0256	0.0000	1.0000
Net profit before tax/Paid-in capital	0.1827	0.0308	0.0000	1.0000
Per Share Net profit before tax (Yuan ¥)	0.1844	0.0332	0.0000	1.0000

3.5 Feature–Target Correlation Analysis

Pearson correlation coefficients were calculated between the numerical financial indicators and the bankruptcy target. Features were ranked using the absolute magnitude of the correlation, while the signed coefficient is retained to indicate the direction of association.

Rank	Feature	Correlation
1	Net Income to Total Assets	-0.315457
2	ROA(A) before interest and % after tax	-0.282941
3	ROA(B) before interest and depreciation after tax	-0.273051
4	ROA(C) before interest and depreciation before interest	-0.260807
5	Net worth/Assets	-0.250161
6	Debt ratio %	+0.250161
7	Persistent EPS in the Last Four Seasons	-0.219560
8	Retained Earnings to Total Assets	-0.217779
9	Net profit before tax/Paid-in capital	-0.207857
10	Per Share Net profit before tax (Yuan ¥)	-0.201395
11	Current Liability to Assets	+0.194494
12	Working Capital to Total Assets	-0.193083
13	Net Income to Stockholder's Equity	-0.180987
14	Borrowing dependency	+0.176543
15	Current Liability to Current Assets	+0.171306

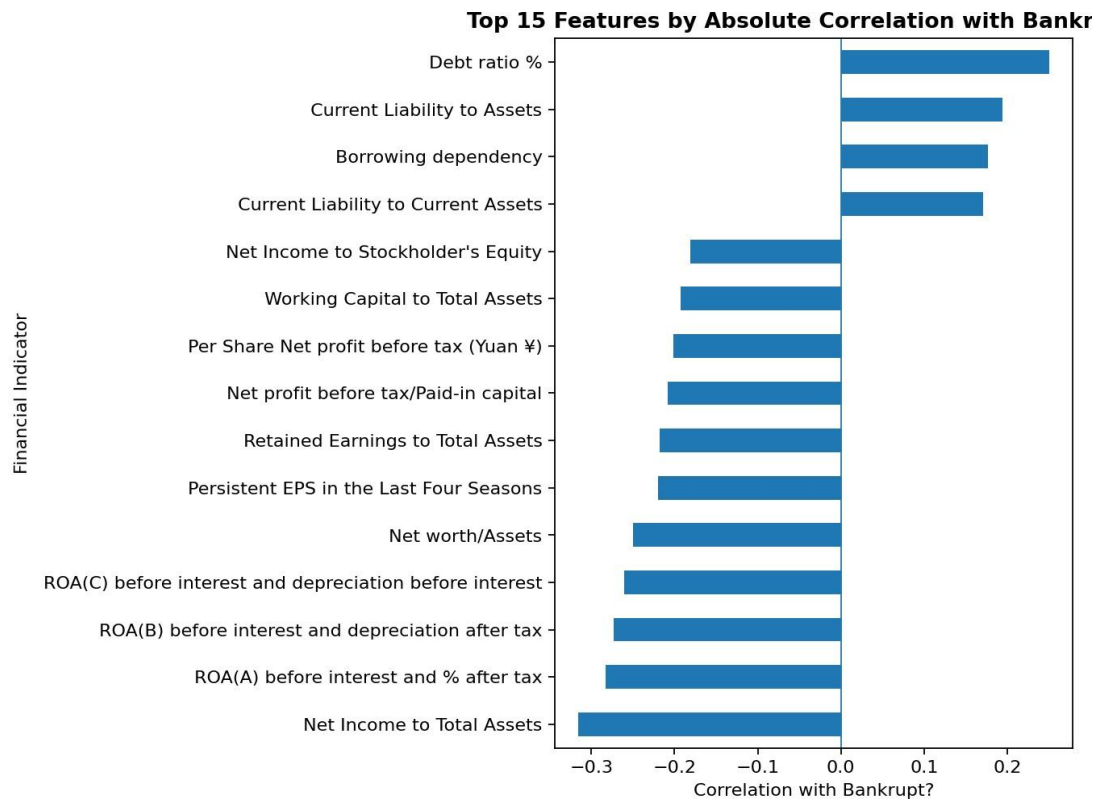


Figure 2. Top 15 financial indicators ranked by absolute correlation with the bankruptcy target.

3.6 Correlation Heatmap

A compact correlation matrix was generated using the fifteen strongest bankruptcy-related features and the target. The heatmap helps identify both feature–target relationships and correlations among the predictors themselves.

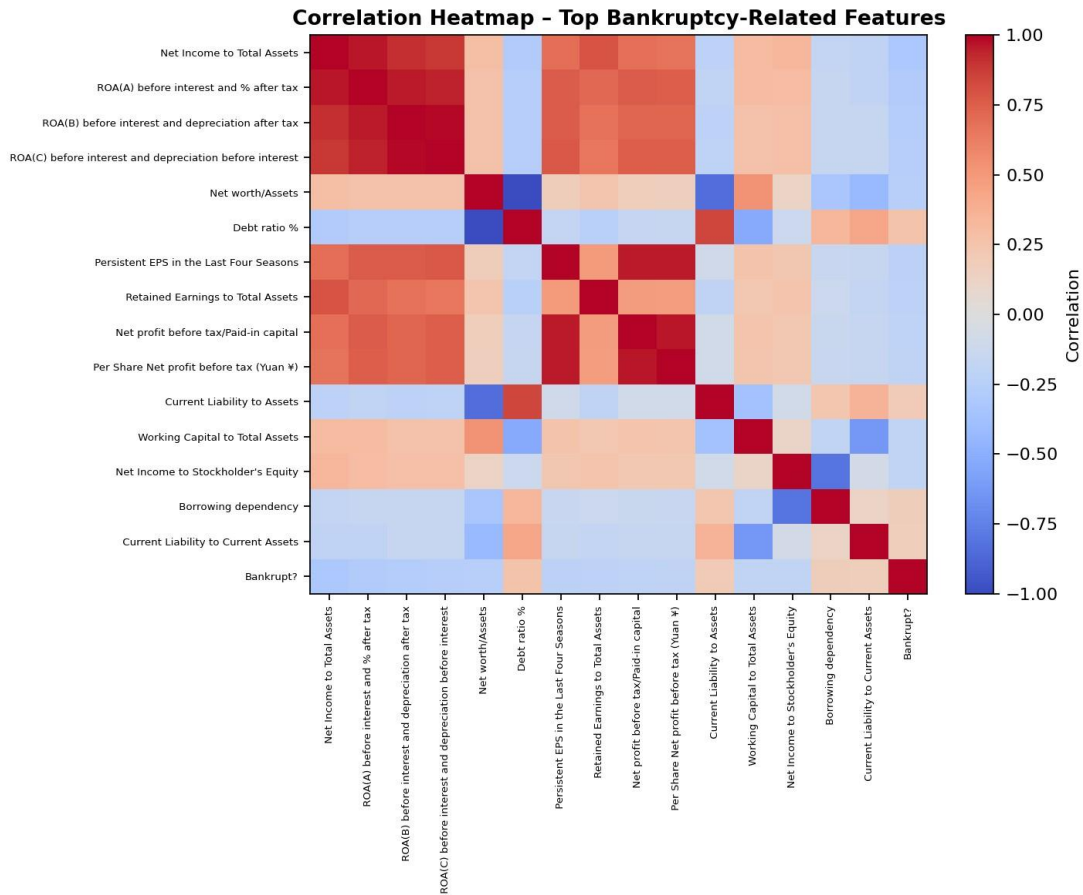


Figure 3. Correlation heatmap for the top bankruptcy-related indicators.

The heatmap shows that several profitability and asset-return indicators have negative relationships with the bankruptcy target, whereas leverage-related indicators such as Debt ratio % and Borrowing dependency show positive relationships. Correlation indicates association, not causation, and should not be interpreted as proof that a feature independently causes bankruptcy.

3.7 Class-wise Boxplot Analysis

The four features with the largest absolute correlations were compared between non-bankrupt and bankrupt companies using boxplots. This provides a visual view of differences in the distributions of important indicators across the two target classes.

Distribution of the Four Strongest Correlated Indicators

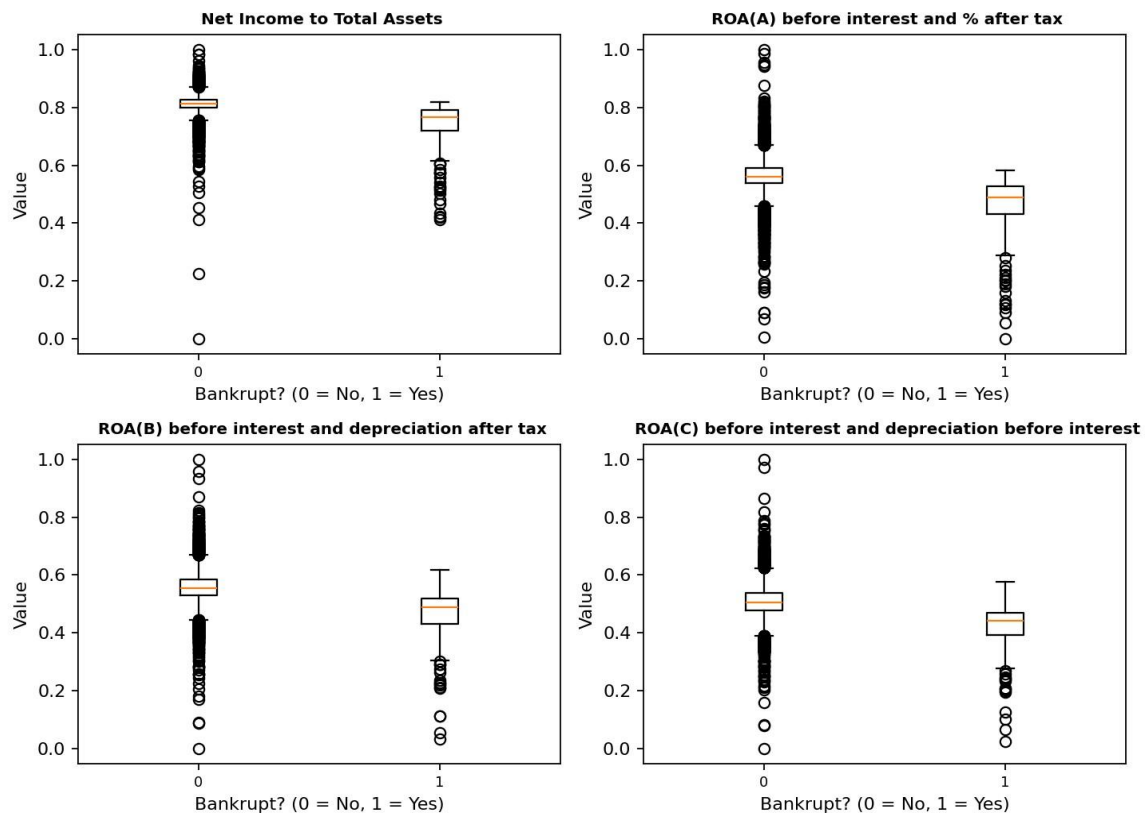


Figure 4. Boxplots comparing the four strongest correlated financial indicators by bankruptcy class.

3.8 Class-wise Mean Comparison

To complement the boxplots, class-wise means were calculated for the ten strongest correlated features. These values are descriptive comparisons and are not used alone to determine model importance.

Feature	Mean: Class 0	Mean: Class 1
Net Income to Total Assets	0.8101	0.7381
ROA(A) before interest and % after tax	0.5620	0.4569
ROA(B) before interest and depreciation after tax	0.5567	0.4615
ROA(C) before interest and depreciation before interest	0.5081	0.4185
Net worth/Assets	0.8893	0.8130
Debt ratio %	0.1107	0.1870
Persistent EPS in the Last Four Seasons	0.2301	0.1888
Retained Earnings to Total Assets	0.9357	0.9042
Net profit before tax/Paid-in capital	0.1839	0.1477
Per Share Net profit before tax (Yuan ¥)	0.1856	0.1478

3.9 Key EDA Findings

1. The dataset contains 6,819 observations, 95 financial features and one binary target.
2. No missing values were found in the project CSV, consistent with the official UCI documentation.
3. No duplicate rows were detected.

4. The feature “ Net Income Flag” is constant and therefore non-informative for predictive modeling.
5. The target is highly imbalanced: 220 bankrupt observations versus 6,599 non-bankrupt observations.
6. “Net Income to Total Assets” has the strongest absolute correlation with the bankruptcy target in this EDA.
7. Several profitability and return-on-assets indicators have negative correlations with bankruptcy, while selected leverage/liability indicators have positive correlations.
8. Class-wise boxplots show distributional differences in important financial indicators between bankrupt and non-bankrupt companies.

CHAPTER 4 – DISCUSSION AND IMPLICATIONS

4.1 Class Imbalance

The class distribution is strongly skewed toward non-bankrupt companies. Only 3.23% of the observations belong to the bankrupt class. This means that accuracy alone is not an appropriate primary measure of model usefulness. The modeling stage therefore uses class-aware training and evaluates Precision, Recall, F1-score, confusion matrices, Precision–Recall curves and Average Precision.

4.2 Financial Indicators

The strongest EDA associations are concentrated around profitability, returns on assets, net worth, retained earnings, leverage and liquidity. In particular, Net Income to Total Assets has the largest absolute correlation with the target. These relationships support the use of financial indicators as predictors, while also showing that multiple indicators should be considered together.

4.3 Implications for Machine Learning

1. The constant feature should not contribute useful predictive information and can be removed.
2. The absence of missing values reduces the need for imputation in the raw dataset, although the project preprocessing pipeline remains capable of handling missing or infinite values.
3. The severe class imbalance motivates the use of class weights and stratified train/test splitting.
4. Decision Tree provides interpretable threshold-based rules but must be controlled to reduce overfitting.
5. Random Forest combines multiple trees and can provide stronger generalisation and feature-importance estimates.
6. Precision–Recall analysis is particularly relevant because the bankrupt class is the minority class and missed bankrupt companies are important errors.

4.4 Limitations of the EDA

Correlation measures linear association and does not establish causality. Some financial variables may also be correlated with one another, so correlation rankings should not be treated as a complete feature-selection procedure. In addition, the dataset represents Taiwanese companies and historical conditions from 1999–2009, so findings may not automatically generalize to every country, industry or current economic environment.

CHAPTER 5 – CONCLUSION

The exploratory analysis establishes a clear understanding of the dataset used for the bankruptcy prediction project. The dataset is a 6,819-record, 95-feature financial classification dataset with a binary bankruptcy target. The project CSV contains no missing values and no duplicate rows, while one constant feature was identified.

The most important EDA observation is the strong class imbalance: 220 bankrupt companies compared with 6,599 non-bankrupt companies. Correlation analysis identifies profitability, asset-return, net-worth, leverage and liquidity indicators as important associations with the target. Visual analysis through the class distribution chart, correlation plot, heatmap and boxplots supports these observations.

These findings provide the foundation for the next stage of the project, where Decision Tree and Random Forest classifiers are trained and compared. Because of the imbalance, the final modeling stage should emphasize Precision, Recall, F1-score, confusion matrices and Precision–Recall analysis rather than relying on accuracy alone.

REFERENCES

1. UCI Machine Learning Repository. Taiwanese Bankruptcy Prediction. Dataset ID 572. DOI: 10.24432/C5004D.
2. Project presentation: “BANKRUPTCY PREDICTION – Using Decision Trees and Random Forest with Precision–Recall Analysis”, A.Y. 2026–27.
3. Project implementation: `bankruptcy_prediction_complete.py`.

Report generated from the project dataset and EDA workflow on 06 October 2026.